

Generative AI in higher education: Ethical and behavioral factors influencing students' intentions to use ChatGPT

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ABSTRACT

This study investigates the behavioral and ethical considerations shaping students' adoption of ChatGPT for academic tasks, emphasizing the broader implications of Generative AI in higher education. By integrating the Fairness, Accountability, Transparency & Ethics in AI (FATE), the Technology Acceptance Model (TAM), the Theory of Planned Behavior (TPB), and the Unified Theory of Acceptance and Use of Technology (UTAUT), we developed and tested an integrated model using survey data from 344 students across four European countries, Estonia, Germany, Poland, and Romania, selected to capture diverse educational and digital contexts in Northern, Western, Central, and Eastern Europe. Our findings highlight that while ethical concerns shape trust, social influence and students' perceived control over their ability to use ChatGPT are stronger predictors of actual usage. We revealed that explainability and privacy are the strongest predictors of trust in ChatGPT, yet trust does not directly influence students' intention to use it. Instead, trust significantly impacts perceived performance, suggesting that students evaluate ChatGPT's reliability through its effectiveness rather than as a factor in adoption. From a behavioral perspective, social influence from university professors and perceived behavioral control emerges as key drivers of ChatGPT adoption. Finally, this research provides practical insights for educators and policymakers on promoting the adoption of generative AI tools in higher education by addressing opportunities, considering associated risks, and determining typical usage patterns among students.

1. Introduction

Today, Artificial Intelligence (AI) is no longer a novelty in higher education, with research on the subject spanning several decades [1]. Lately, the adoption of AI tools, particularly generative models, has accelerated at an unprecedented pace [2,3]. Recent advances in Generative AI (GenAI), especially ChatGPT – currently the most widely used GenAI system in academic settings and the fastest-growing app in history [4] – are demonstrating groundbreaking results, such as successfully passing complex exams in fields such as law, business, or medicine [5].

This rapid advancement has sparked discussions about the ethical, behavioral, and policy-related implications of GenAI in education [6,7]. As universities experiment with GenAI integration, concerns about fairness, privacy, transparency, and accountability have moved to the

forefront of academic and policy debates [8,9]. In light of these discussions, and the increasing popularity of AI tools, understanding students' adoption of GenAI and its ethical implications has become a key concern for higher education institutions.

Unlike many earlier digital tools in education, GenAI systems such as ChatGPT exhibit anthropomorphic characteristics, including conversational ability and human-like responses. These qualities may lower barriers to adoption for students by making interactions more intuitive and engaging [10]. At the same time, anthropomorphism does not necessarily mitigate skepticism among educators, who may remain concerned about the tool's reliability, educational value, and alignment with pedagogical objectives [11,12]. Recent studies show that while students tend to view GenAI as a productivity enhancer, faculty perspectives are more cautious due to risks such as hallucinations, fabricated citations, and erosion of critical thinking [13,14].

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Beyond student adoption, recent research highlights that GenAI tools are also being employed in the design of curricula and instructional materials. Studies demonstrate how Large Language Models-augmented frameworks can support curriculum innovation [15,16] and provide empirical insights into both the opportunities and constraints of AI-driven curriculum development [16]. This broader perspective indicates that Generative AI is not only reshaping students' learning practices but also transforming institutional approaches to teaching and curriculum design, going beyond the hype.

As demonstrated by the growing number and areas of research [17], the field of AI has expanded significantly over time. Yet higher education systems continue to struggle to keep pace with rapid AI innovation, leading to gaps between technological developments and pedagogical preparedness [18]. Consequently, many problems and opportunities remain unresolved or unexplored [19]. Existing research focuses primarily on theoretical models, often lacking empirical validation of AI adoption in educational settings Memarian & Doleck [20]. As one might assume, the adoption and use of ChatGPT in university activities offer tangible benefits, including an improved learning experience, faster and better completion of university assignments, and accessibility and ease of use [21]. As *technology* continues to advance, universities are increasingly recognizing the importance of integrating AI-powered tools such as ChatGPT into their educational strategies [22]. However, due to the novelty of the technology, student feedback on these aspects is rather scarce [23].

One of the key challenges in AI adoption is *trust* and *reliability*. In this respect, the terms "Fairness", "Accountability", "Transparency", and "Ethics" (FATE) have gained popularity in AI discourse, particularly in higher education [24]. Privacy, ethical concerns, and behavioral control have also been widely discussed in the context of AI adoption [25]. Ensuring that ChatGPT is used responsibly in education requires a clear understanding of students' perceptions and behavioral tendencies [20]. To maximize its potential while maintaining ethics, users should be aware of its limitations, apply critical thinking, double-check information, and stay tuned for updates and safety precautions. This aligns with The Global Education 2030 Agenda published by UNESCO, which emphasizes the need for human-centered AI policies to ensure ethical, safe, and equitable use of GenAI in education [4]. Furthermore, to ensure that GenAI serves as an effective and responsible complement to traditional education, it is essential to analyze students' *behavioral* aspects when employing ChatGPT in academic settings. This can enable academic institutions to successfully adjust, direct, control and monitor the use of AI for the benefit of students and the academic community. The behavioral aspects of students using ChatGPT can be examined through various psychological factors, such as social norms, subjective norms, social influence, and intention to use AI-based tools [26]. Universities can play a significant role in shaping these behaviors by establishing clear guidelines, promoting responsible AI use, and fostering a learning environment that supports ethical AI integration.

To sum up, recent research [27] calls for a new educational philosophy that accommodates the integration of GenAI in higher education. However, ensuring smooth and responsible AI adoption requires a deeper understanding of students' attitudes toward such tools, as well as their *ethical* and *behavioral* considerations regarding AI use in academia. Despite growing scholarly attention, empirical studies that integrate technological, behavioral, and ethical predictors of ChatGPT adoption remain limited, especially across different national contexts [28,29]. To the best of our knowledge, as of April 2024, only one study has synthesized multiple theoretical models to examine AI chatbot adoption in higher education. This study proposed a research framework that integrates elements from TAM, TRI 2.0, UTAUT2, revised TAM, and TPB, incorporating variables such as optimism, innovativeness, discomfort, and insecurity [30].

Despite the increasing attention, current research on ChatGPT adoption in higher education reveals three critical gaps. *First*, existing studies typically apply individual theoretical models such as TAM

[31–34], TPB [35], or UTAUT [7,26], but rarely integrate them with ethical frameworks like FATE, leaving unexplored the intersection of behavioral and ethical dimensions even though research on AI in education highlights that fairness, accountability, transparency, and privacy are essential for responsible use [20,36,37]. As a result, it remains unclear how students weigh functional benefits against ethical risks when evaluating ChatGPT. *Second*, prior studies on ChatGPT adoption in higher education are often limited to single-country contexts [32–34,38] or remain primarily conceptual or review-based [39–42], with cross-national empirical work remaining relatively rare. This restricts the generalizability of findings and obscures whether adoption drivers are culturally robust. *Third*, the role of ethical considerations such as fairness, accountability, transparency, and privacy in shaping trust and influencing adoption decisions remains underexamined, despite evidence that trust is central to technology acceptance [43,44] and is influenced by ethical and explainability factors [45,46].

To address these gaps, this study offers several distinct contributions to the literature on GenAI in higher education. First, we develop and empirically validate an integrated model that combines the Technology Acceptance Model (TAM), the Theory of Planned Behavior (TPB), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the FATE framework, thereby linking detailed ethical constructs (explainability, transparency, accountability, fairness, privacy and trust) with behavioural and technological determinants of adoption in a single framework. Second, using survey data from 344 students in four European countries (Estonia, Germany, Poland and Romania), we test this model with PLS-SEM and show that it explains substantial variance in intention to use and intention to purchase ChatGPT, providing cross-national evidence on adoption patterns in diverse educational and digitalisation contexts. Third, we demonstrate that ethical perceptions strongly shape trust and perceived performance, but that, contrary to much prior technology-acceptance work, trust does not directly predict intention to use once usefulness and attitudes are taken into account; instead, trust operates indirectly through performance evaluations, refining current theorisation of human trust, AI interaction.

The paper is structured as follows: [Section 2](#) forms the theoretical background, [Section 3](#) describes the methodology, [Section 4](#) provides an analysis of the results obtained, and [Section 5](#) considers the implications, limitations and future work, while [Section 6](#) concludes the paper.

2. Background

2.1. Artificial intelligence in education

AI represents a technology with capacity to automate tasks and process large quantities of data [47] to provide human-like responses [48]. Recent advances in AI, particularly in natural language processing and generative models, have rapidly expanded interest in how AI can be adopted in education and how its adoption can be explained [39,41,49,50].

AI automates tasks, processes vast amounts of data [47], and generates human-like responses [48]. Among GenAI tools, ChatGPT is the most recognized, with the potential to revolutionize education by enhancing personalized learning, engagement, and [51]. It allows educators to tailor lessons and students to focus on specific learning needs [41]. However, educators remain hesitant to fully embrace it due to uncertainties about its pedagogical implications [52]. While ChatGPT offers sophisticated natural language processing and personalized real-time responses, its weaknesses include limited human interaction, data dependency, biases, and ethical concerns [39,40]. These risks pose challenges such as threats to academic integrity, reduced higher-order cognitive skills, and misinformation. Despite these concerns, ChatGPT enhances accessibility, facilitates complex learning, and reduces educators' workload. The potential benefits highlight the need for responsible AI integration to maximize advantages while mitigating risks [40].

Recent studies have explored AI adoption in education, examining

student and teacher perceptions while addressing ethical concerns such as misinformation and copyright issues [42,53,54]. Research indicates that factors influencing ChatGPT adoption among students include ease of use, perceived usefulness, social influence, and behavioral/cognitive aspects [31]. For educators, ChatGPT adoption is linked to time-saving opportunities, academic self-efficacy, and stress reduction but is negatively impacted by peer influence and academic integrity concerns [55]. Nevertheless, resistance persists, largely due to doubts about its educational value and difficulties in adapting assessment methods, with concerns extending beyond simple technology aversion to a perceived cost–benefit imbalance. This underlines the need for comprehensive GenAI literacy and institutional support to promote responsible integration [11].

Concerns regarding ChatGPT's reliability and trustworthiness have also been raised, as studies have documented inaccuracies and fabricated references in its outputs Cooper [56]. As a result, educators must critically assess AI-generated materials to ensure credibility and adapt them for pedagogical use [56–58]. While public perception of AI in education is largely positive, concerns such as cheating, privacy, and content accuracy must be addressed to ensure ethical integration. Researchers emphasize the need for institutional guidelines to safeguard responsible AI use in academia [59].

Given AI's transformative role in education, structured frameworks are essential for understanding adoption patterns and ethical implications. Technology adoption models provide insights into the factors shaping ChatGPT's integration into learning environments. Most studies employ single-framework approaches, predominantly UTAUT [7,38] and TAM [32,34], TPB [35]. Alternative models such as SCT (Social Cognitive Theory) [55], TCS (Technology Continuance Theory) [60], Innovation Resistance Theory [57], and TOEK [61] have also been used. However, reliance on a single model limits understanding of the complex relationships between technological, behavioral, and ethical dimensions. However, reliance on a single model limits the ability to account for the multidimensional nature of GenAI adoption, which involves not only technological perceptions but also ethical concerns and behavioral norms [62]. As a result, more comprehensive models are necessary to capture the interplay between technological, behavioral, and ethical factors.

To bridge this gap, we seek to develop an integrated technology adoption model specifically tailored to the context of GenAI adoption in academia. This model combines multiple theoretical perspectives to provide a holistic analysis of the ethical, behavioral, and technological aspects of GenAI usage. Section 2.2 examines theoretical models relevant to our approach, alongside reflections on previous studies that have applied these models in similar contexts.

2.2. Technology adoption models

The study of technology adoption is currently being built on several *technology adoption* models. The Technology Acceptance Model (TAM) and its extensions have been widely used to study the adoption of new technologies in different domains, including education [63,64]. TAM, developed by Fred Davis [65], declares that perceived usefulness – the degree to which a user believes a system would enhance their job performance – and perceived ease of use – the degree to which a user believes a system would be free of effort – are primary determinants of users' attitudes and intentions with regard to using technology. Consequently, TAM is likely to have a narrow focus on technology adoption and usage behaviors, particularly in organizational contexts [66]. As such, it may not fully capture the broader range of factors influencing human behavior in non-technological contexts. In the context of education, researchers have applied TAM to understanding educators' and students' attitudes towards the use of AI tools, and their willingness to incorporate these technologies into teaching and learning processes [63]. Thus, despite its widespread use, TAM has been critiqued for its narrow focus, limited ability to account for social and contextual

influences, and reduced explanatory power in complex environments such as AI-supported learning.

The Unified Theory of Acceptance and Use of Technology (UTAUT) [67] expands TAM by integrating constructs such as performance expectancy, effort expectancy, social influence, and facilitating conditions. UTAUT has been applied in educational contexts to explore adoption patterns among both students and educators [68].

However, as recent work suggests, UTAUT alone cannot fully explain GenAI adoption because it does not incorporate ethical concerns or trust factors that are central to AI decision-making [69].

Ethical considerations are increasingly recognized as pivotal in AI adoption [70]. The FATE model, an acronym for Fairness, Accountability, Transparency, and Ethics, was proposed by Dignum [71], provides a structured framework for addressing ethical considerations in the development and deployment of AI systems.

Another prominent model in the context of education and technology adoption is the Theory of Planned Behavior (TPB), introduced by Ajzen [72] and used to study technology acceptance behavior [73]. According to TPB, behavioral intentions are influenced by three primary factors: attitude towards the behavior, subjective norms, and perceived behavioral control. TPB does not specifically address technology adoption but rather provides a framework for understanding human behavior in various domains, including education. It is often criticized for its simplistic assumptions and neglect of external factors. As TAM and TPB address different aspects, they are often used in combination, such as in studies on the adoption of learning technologies [74].

Thus, each model offers valuable but partial insights. Importantly, none of these frameworks alone adequately capture the convergence of technological perceptions, ethical risks, trust formation, and social influences inherent in GenAI adoption. Therefore, our research introduces an integrated research model that unites three perspectives – *ethical* (FATE), *behavioral* (TPB) and *technological* (TAM and UTAUT) – to comprehensively examine students' intention to adopt and use ChatGPT for university assignments. This model will be tested within the education context in four countries (Estonia, Germany, Poland, and Romania). To this end, we formulated two research questions (RQ) to guide our study:

RQ1: *What ethical, behavioral and technological factors influence students' attitudes towards using ChatGPT for university assignments, and to what extent?*

RQ2: *What challenges are associated with the effective use of ChatGPT for university assignments and the ethical integration of ChatGPT into academia?*

2.3. Integrated research model and hypotheses

Building on the models discussed above, we integrate technological, behavioral, and ethical perspectives into a single research model. Specifically, it builds on the TAM [75] and incorporates extensions from revised versions of TAM and the UTAUT [43,67,76] to explain students' adoption of ChatGPT. To capture behavioral dimensions, we draw on the TPB, which emphasizes attitudes toward unethical ICT behavior [23, 77], perceived behavioral control [78], and subjective norms (social influence from peers and professors) [79,80]. Finally, ethical concerns are incorporated using the FATE framework [10,46], which we operationalize through accountability, explainability, fairness, privacy, trust, and transparency [79,81,82]. This integrated approach allows us to capture the interplay of technological, behavioral, and ethical drivers of ChatGPT adoption in higher education.

2.3.1. Ethical aspects

Ethical constructs such as explainability, transparency, accountability, fairness, privacy, and trust form the core of responsible AI adoption in public-facing contexts. These constructs are grounded in several theoretical frameworks, including trust theory, fairness–justice theory, privacy calculus, and technology acceptance models (e.g., TAM,

UTAUT). Together, these frameworks explain how users evaluate AI systems when consequences, uncertainty, and asymmetry of information are high.

Explainability is widely considered a foundational requirement for fostering trust in AI systems. When users understand how model outputs are generated, they perceive the system as more predictable, controllable, and aligned with their expectations [45,83]. Explainability reduces the perceived “black box” nature of machine learning models, lowering uncertainty and cognitive resistance [84,85]. Empirical studies in human - AI interaction further show that interpretable explanations mitigate perceived risks and strengthen users’ confidence in AI-supported decisions [86]. Taken together, these insights motivate us to propose the following hypothesis:

H1: Higher levels of *explainability* positively influence *trust*.

Transparency, defined as disclosing inputs, processes, and decision logic, is a precursor to explainability because it provides the necessary information for users to understand system behaviour [87,88]. Transparent AI systems reveal model constraints, training data characteristics, and reasoning pathways, enabling more meaningful explanatory outputs [89,90]. Prior work demonstrates that transparency enhances cognitive comprehension of AI models and allows users to critically engage with explanations rather than passively accept outputs [83,91]. Accordingly, we hypothesize that:

H2: Higher levels of *transparency* positively influence *explainability*.

Accountability mechanisms such as traceability, auditability, and responsible oversight - ensure that algorithmic decisions can be scrutinized and justified, which strengthens perceptions of fairness [92,93]. In algorithmic governance, accountability reduces arbitrariness and supports equitable treatment by clarifying who is responsible for decisions and how errors are addressed [94,95]. Research shows that accountability frameworks enhance fairness judgments by reducing perceived bias and clarifying the distribution of decision authority [96,97]. Following this reasoning, we hypothesize:

H3: Higher levels of *accountability* positively influence *fairness*.

Accountability requires that AI decisions can be justified, which inherently demands explainability [87,90]. Systems lacking explanation mechanisms cannot meet accountability standards such as transparency, rational justification, and traceable decision-making [93,95]. Empirical studies show that accountability pressures lead organizations to build more interpretable AI systems, improving users’ understanding and reducing perceived opacity [83,85]. Thus, we propose the following hypothesis:

H4: Higher levels of *accountability* positively influence *explainability*.

Fairness perceptions strongly drive trust in algorithmic systems. When users perceive AI outputs as bias-free, equitable, and socially just, they are more willing to rely on and adopt the system [96]. Prior studies show that perceived algorithmic fairness reduces concerns about discrimination and moral harm, which in turn enhances trust and acceptance [98]. Fairness is particularly salient in public-sector contexts, where legitimacy depends on equal treatment and procedural justice [97,99]. Accordingly, we hypothesize:

H5: Higher levels of *fairness* positively influence *trust*.

Privacy protection is a foundational antecedent of trust in digital technologies. Research shows that perceived privacy safeguards reduce fears of misuse, surveillance, or unauthorized data processing, thereby fostering higher trust [100]. In AI contexts, privacy-preserving systems signal responsible design and ethical commitment, increasing users’ willingness to engage with algorithmic tools [101]. Empirical findings consistently reveal that privacy assurances positively influence trust and adoption intention across digital platforms [102,103]. Hence, we posit the following hypothesis:

H6: Higher levels of *privacy* positively influence *trust*.

Transparency enhances fairness perceptions by revealing how decisions are made and enabling users to identify potential biases [87,88]. Users perceive transparent systems as more procedurally just because information disclosure improves accountability and reduces suspicions

of hidden discrimination [89,99]. Prior research shows that transparency increases both distributive and procedural fairness judgments by enabling scrutiny of algorithmic processes [96,98]. Therefore, we propose the following hypothesis:

H7: Higher levels of *transparency* positively influence *fairness*.

Trust is a key determinant of perceived performance in technology acceptance research. Users who trust a system judge its outcomes as more accurate, reliable, and useful [76]. Trust reduces uncertainty and cognitive effort, leading users to interpret system performance more favourably [86]. In AI contexts, trust has been shown to enhance perceived predictive value, efficacy, and task support [104]. Based on this evidence, we hypothesize:

H8: Higher levels of *trust* positively influence perceived *performance*.

Trust is one of the strongest predictors of behavioural intention in TAM, UTAUT, and related adoption models [75,105]. When users trust AI systems, they perceive lower risks and exhibit higher willingness to rely on them in decision-making (Dwivedi et al., 2023). Trust supports both initial acceptance and sustained use, especially in contexts involving uncertainty, complexity, or algorithmic opacity [86,104]. Thus, we put forward the following hypothesis:

H9: Higher levels of *trust* positively influence intention to *use*.

2.3.2. Behavioral aspects

Within the TPB, perceived behavioural control reflects individuals’ confidence in their ability to perform a behaviour and is one of the strongest predictors of intention [72,106]. When students believe they have sufficient skills and resources to use AI tools, their likelihood of adopting such tools increases. Prior research in digital learning environments shows that technological self-efficacy significantly enhances intention to use educational technologies, including AI-assisted platforms [78,107]. Recent studies focusing specifically on ChatGPT and GenAI similarly demonstrate that perceived competence in navigating AI systems strongly predicts adoption intentions [12,108]. Accordingly, we hypothesize:

H10: Perceived *behavioral control* positively influences intention to *use*.

Subjective norms - social pressures perceived from relevant others - constitute a central determinant of intention within TPB [72]. Peer influence is particularly salient in academic environments, where students learn from one another, observe shared practices, and adopt technologies normalized within their social group. Studies on AI and educational technologies show that peer endorsement or usage patterns can significantly increase students’ openness to adopting ChatGPT and similar tools [107,108]. In peer communities, normative pressures operate both explicitly (recommendations, encouragement) and implicitly (observing others’ successful use), heightening adoption likelihood [11]. Thus, we propose:

H11: *Social norms* exerted by colleagues positively influence intention to *use*.

In academic settings, instructors hold substantial normative authority and can legitimize or discourage emerging technologies through explicit guidance, classroom rules, and modelling of appropriate use. Research shows that professor-endorsed norms strongly influence students’ uptake of digital tools because instructors are viewed as credible, expert, and institutionally sanctioned role models [11]. Studies on AI adoption in universities confirm that when professors encourage or allow the use of ChatGPT, students demonstrate significantly higher behavioural intention - more so than in response to peer influence alone [12,107]. To formally state this expectation, we hypothesize:

H12: *Social norms* exerted by professors positively influence intention to *purchase*.

Beyond influencing everyday academic practices, professors’ attitudes may also shape students’ willingness to invest financially in technology. When instructors frame AI tools as academically valuable, essential for skill development, or professionally relevant, students may perceive personal ownership (or paid versions) as worthwhile [11]. Prior work on educational technology adoption shows that institutional

endorsement increases not only free usage but also willingness to pay for premium versions that offer enhanced functionality [80]. Therefore, we hypothesize:

H13: *Social norms* exerted by professors positively influence *intention to use*.

2.3.3. Technology adoption and use aspects

The TAM and UTAUT posit that system characteristics influence user beliefs, particularly ease of use [75]. When users perceive a system as performing reliably and efficiently, they experience fewer barriers, cognitive demands, and operational frictions - conditions that directly enhance perceived ease of use [78]. Studies of AI-supported learning tools similarly show that performance expectations shape perceptions of usability [7]. Accordingly, we hypothesize that:

H14: *Performance* positively influences *ease of use*.

System performance has consistently been identified as a determinant of positive affective evaluations, as users develop more favorable attitudes when they believe a technology improves productivity or delivers high-quality outcomes [76]. Recent findings on AI tools in educational contexts demonstrate that perceived performance is strongly associated with more positive attitudes toward adoption [32]. Hence, we theorize that:

H15: *Performance* positively influences *attitude toward use*.

A central proposition of TAM is that ease of use enhances perceived usefulness because technologies requiring less effort are more likely to be judged as capable of improving task performance [75]. Empirical studies on GenAI and digital learning systems confirm this relationship, demonstrating that ease of use enhances evaluations of usefulness [34]. Thus, we propose that:

H16: *Ease of use* positively influences *usefulness*.

Ease of use also shapes attitudes directly. When users perceive a system as intuitive and effortless, they form more positive emotional responses that facilitate intention formation [78]. Evidence from recent research on ChatGPT adoption similarly indicates that ease of use is positively associated with favorable attitudes [7]. Therefore, we hypothesize that:

H17: *Ease of use* positively influences *attitude toward use*.

Perceived usefulness is one of the strongest predictors of attitudes toward technology [76]. In both classical and contemporary studies, including research on AI-based systems, users who believe that a system enhances their performance exhibit significantly more positive attitudes [32]. Accordingly, we theorize that:

H18: *Usefulness* positively influences *attitude toward use*.

Usefulness is also a well-established antecedent of behavioral intention [67,75]. When users perceive a technology as offering tangible functional benefits, they demonstrate a stronger intention to use it. Findings from recent empirical studies reinforce this relationship in the context of ChatGPT and AI-based learning tools [34]. Thus, we hypothesize that:

H19: *Usefulness* positively influences *intention to use*.

TAM and UTAUT research repeatedly show that attitude toward using a technology is a proximate determinant of behavioral intention [76]. Positive evaluations of AI systems, such as trust in their capabilities and satisfaction with their operation, substantially strengthen usage intentions [7]. Therefore, we propose that:

H20: *Attitude toward use* positively influences *intention to use*.

Finally, intention to use has been shown to precede more advanced behavioral commitments such as intention to purchase or subscribe, as demonstrated in technology continuance research [109]. When users expect continued value and perceive a system as beneficial, they are more willing to invest financially. Accordingly, we hypothesize that:

H21: *Intention to use* positively influences *intention to purchase*.

2.3.4. Moderating effects

Technology adoption research highlights that the effects of core TAM constructs are not uniform across contexts but often depend on ethical

perceptions and socio-institutional environments. Ethical considerations, such as perceived fairness, transparency, and responsible use, have been shown to shape user evaluations of usefulness and their willingness to apply AI-based tools [46,110]. When users view AI as ethically aligned, the positive influence of usefulness on intention is strengthened; when ethical concerns are salient, this relationship weakens. Recent studies on GenAI adoption similarly show that ethical risk perception moderates adoption pathways, particularly those involving usefulness and trust (Zhang et al., 2023; Naeem, 2023). Accordingly, we hypothesize that:

H22: The interaction between *ethics* and *usefulness* moderates the relationship between usefulness and intention to use.

Cross-national research demonstrates that institutional context, cultural values, and regulatory environments significantly moderate technology adoption processes ([111]; Dwivedi et al., 2023). In particular, the link between intention to use and intention to purchase is sensitive to structural conditions such as digital readiness, economic constraints, and risk perceptions across countries [68,112]. GenAI studies further suggest that willingness to financially commit to AI tools varies widely across regions [113]. Thus, we theorize that:

H23: The interaction between *country* and *intention to use* moderates the relationship between intention to use and intention to purchase.

As such, the research model integrates technological, behavioral, and ethical perspectives into a single research model provided as Fig. 1.

3. Methodology

To empirically test the integrated research model presented in Section 2.3, we designed a survey instrument drawing on validated scales from prior literature. The constructs from TAM, TPB, UTAUT, and FATE were adapted to the context of students' adoption of ChatGPT in higher education and operationalized into a structured measurement instrument, resulting in a list of 17 constructs (Table 1). The questionnaire was organized into five sections.

The first section collects demographic information and experiences with ChatGPT in academia through 8 questions (Appendix A¹, Part 1). The second section measures students' adoption and use intentions (Appendix A, Part 2). The third section focuses on evaluating students' perceptions of ChatGPT's ethics (Appendix A, Part 3). The fourth section aims to explore the behavioral aspects of students' use of ChatGPT with regard to academic activities (Appendix A, Part 4). Lastly, the fifth section seeks to elucidate students' motivations, adoption, and purposes for utilizing GenAI tools in addition to ChatGPT (Appendix A, Part 5). The predefined options were based on the list of the most competitive GenAI tools found in news, forums, and reports.

Each measurement item was measured on a five-point Likert scale ranging from "strongly disagree" to "strongly agree." Where possible, items were adapted from previously validated studies to ensure reliability and contextualized to the academic setting. For example, constructs relating to fairness, accountability, and transparency were adapted from Shin [10,46] and Gupta et al [82], while items capturing perceived behavioral control were drawn from Venkatesh & Bala [78]. Social influence constructs distinguished between peers and professors, following Gonçalves et al [79] and Kim et al [80] (see Table 1).

A pilot study was conducted to test the questionnaire in terms of reliability. At first, five academic experts checked its consistency, sequence and relevance of the measurement items. The questionnaire was then shared among one academic group per country in September 2023. To recruit participants, we applied a non-probability convenience sampling strategy, complemented with purposive targeting to ensure inclusion of higher education students with experience in ChatGPT or other GenAI tools. The survey was distributed in October–November 2023 through multiple channels, including university mailing lists,

¹ Appendix A

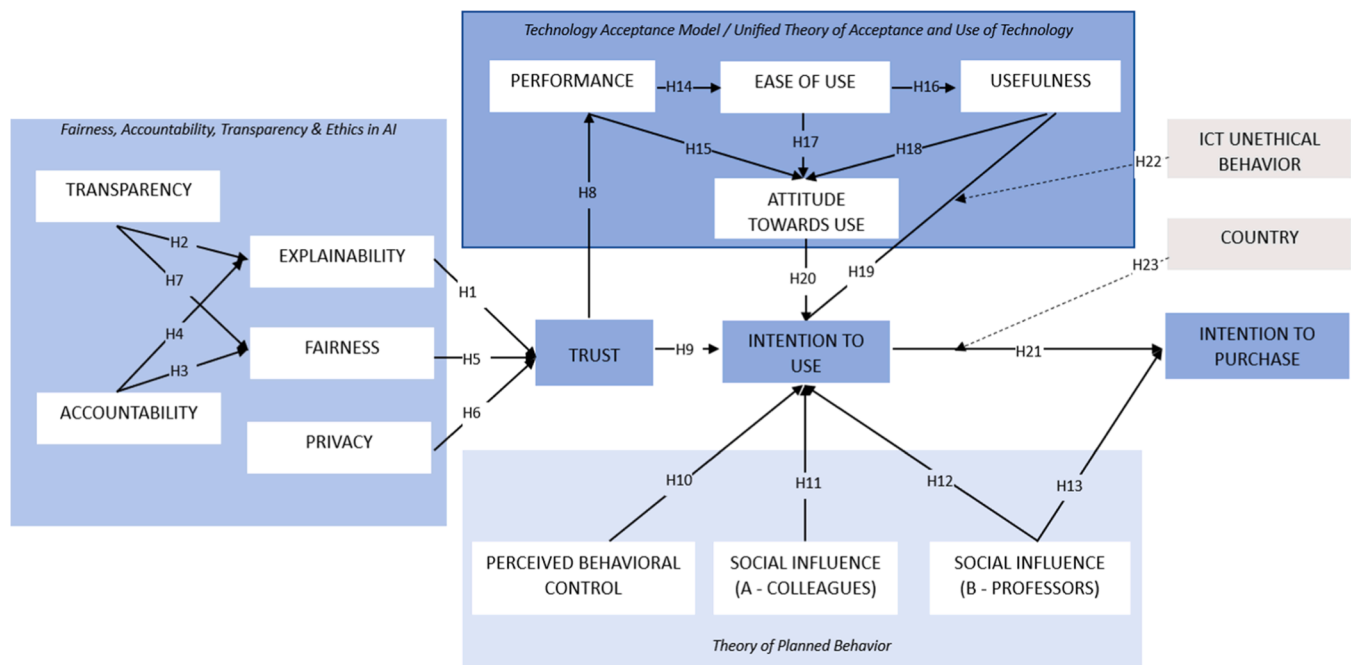


Fig. 1. An integrated model for students' adoption of ChatGPT in university activities.

learning management systems, social media groups, and through faculty members acting as institutional contact points in Estonia, Germany, Poland, and Romania. Participation was voluntary and no incentives were provided. To ensure replicability and data quality, inclusion criteria required that students had prior exposure to ChatGPT or generative AI, while incomplete responses and those indicating no prior use of AI tools were excluded from the final dataset. This procedure resulted in 344 valid responses across the four countries. The selection of the four European countries was driven by the aim to provide a research context that is both diverse and comparable. These countries represent different regions of Europe (Northern, Western, Central, and Eastern), ensuring variation in cultural and educational contexts. Estonia and Germany are recognized for their relatively high levels of digitalization and early adoption of AI tools in education (e.g., Estonian AI leap²), while Poland and Romania provide perspectives from countries where institutional adoption of AI in education is progressing more gradually. This cross-country variation allows us to examine students' adoption of ChatGPT in heterogeneous educational environments and enhances the generalizability of our findings across different European contexts.

The Partial Least Squares Structural Equation Modeling (PLS-SEM) using *SmartPLS* 3.3.9 software [114] was applied to test the higher education students' intentions to use ChatGPT. PLS-SEM was selected for four main reasons. First, the primary goal of our study is prediction-oriented, focusing on maximizing the explained variance in students' adoption of ChatGPT rather than solely on theory confirmation. Second, our research model integrates constructs from TAM, TPB, UTAUT, and FATE, resulting in a relatively complex structure. Third, given our sample size of 344 students and the cross-country nature of the dataset, PLS-SEM provides more statistical power and does not impose strict distributional assumptions, making it more appropriate than covariance-based SEM. Finally, PLS-SEM has been widely applied in educational technology and information systems research when exploring new or extended models, making it well suited to our study [115].

To study each research model, *first*, the reflective measurement model was developed (based on Fig. 1). *Second*, the measurement model

was evaluated to establish its reliability and validity using the following three measures: (1) indicator loadings and internal consistency reliability – Cronbach's alpha (α) and Composite Reliability (CR); (2) convergent validity (average variance extracted, AVE); and (3) discriminant validity. The constructs' discriminant validity was tested using the (i) Fornell-Larker; (ii) cross-loading; and (iii) heterotrait-monotrait (HTMT) criteria. *Third*, we estimated the path coefficients for the structural model; coefficient of determination (R^2) for the relevance of the construct measures [115]. *Fourth*, the absence of multicollinearity was confirmed by calculating the Variance Inflation Factor (VIF) for all predictor variables. The VIF measures the extent to which the standard error of an estimated coefficient is inflated due to collinearity, with values below the recommended threshold of 5.0 indicating no problematic multicollinearity [116].

4. Results

In this section, we present the results of our study. First, we reveal the demographic characteristics of the respondents. The measurement model assessment results are then presented. We highlight the results derived from the structural model assessment. Finally, patterns and challenges that arose when the students use ChatGPT are described.

4.1. Respondents' characteristics

The initial sample consisted of 528 responses from Estonia (EE), Germany (GE), Poland (PL), and Romania (RO). To align our sample with the research criteria, we applied the following filters: (1) only respondents who answered "Yes, for university (-related) assignments only" or "Yes, for both university and private/professional purposes" to the question "Do you use ChatGPT?" were included, resulting in the elimination of 177 responses; (2) respondents over 40 years of age were excluded due to a lack of representativeness within this age group in the overall sample, leading to the removal of eight responses. Consequently, the final sample used for further analysis comprised 344 responses. Males predominated among the respondents from EE, GE, and PL, but in RO, females were more highly represented (79.26%). Overall, the majority of respondents are bachelor's degree students (60.47%), though EE and RO have a predominance of master's degree students (62.26%)

² Estonia launches AI Leap 2025 to transform education - Education Estonia

Table 1
Constructs with items of measurement.

Construct	Item	Statement	Reference
CHATGPT ETHICS (FATE)			
Accountability	ACCOUNT_1	I think that tools like ChatGPT require a person in charge who should be accountable for its adverse individual or societal effects in a timely fashion	[10,46]
	ACCOUNT_2	Tools like ChatGPT should be designed to enable third parties to examine and review the behavior of an algorithm	
	ACCOUNT_3	Tools like ChatGPT should be designed to allow for adjustments or modifications in its behavior and settings to suit one's preferences	
Explainability	EXPLAIN_1	I find ChatGPT to be easily understandable	[10,46]
	EXPLAIN_2	I think ChatGPT services are interpretable	
	EXPLAIN_3	I can figure out the internal mechanics of ChatGPT	
	EXPLAIN_4	I think that the ChatGPT algorithm can be clearly explainable	
Fairness	FAIR_1	ChatGPT has no favoritism and does not discriminate against people (e.g. in terms of race, nationality, gender, age)	[10,46]
	FAIR_2	The source of data throughout ChatGPT and its data sources should be clearly identified, documented, and subject to performance evaluation and comparison	
	FAIR_3	I believe ChatGPT follows due process of impartiality with no prejudice	
Privacy	PRIVACY_1	I believe that I am interacting with secure AI algorithms that are designed to prevent hacking, data breaches, or cyberattacks	[46]
	PRIVACY_2	I believe that interaction with ChatGPT does not lead to distress, emotional harm, or harassment	
	PRIVACY_3	I believe that ChatGPT will take responsibility for the requests they process on my account	
	PRIVACY_4	I believe my personal information and data are protected and there is no unauthorized access or misuse of my data	
Trust	TRUST_1	I generally trust ChatGPT recommendations, but always check their relevance	[46,79,81]
	TRUST_2	I generally trust ChatGPT recommendations, but sometimes check their relevance	

Table 1 (continued)

Construct	Item	Statement	Reference
Transparency	TRUST_3	I fully trust the recommendations by ChatGPT-driven responses	[10,46,82]
	TRANSP_1	I think that the evaluation and the criteria of ChatGPT used should be publicly released and understandable to people	
	TRANSP_2	Any outputs produced by ChatGPT should be explainable to people affected by those outputs	
	TRANSP_3	ChatGPT should let people know how well internal states of algorithms can be understood from knowledge of its external outputs	
BEHAVIORAL ASPECTS OF USING CHATGPT (Theory of Planned Behavior)			
ICT unethical behavior	ETHIC_1	If I were prohibited from using ChatGPT for preparing university (-related) assignments I would never try to use it	[23,77]
	ETHIC_2	If I were prohibited from using ChatGPT for preparing university (-related) assignments, I would never rely on it (even if I would try it)	
	ETHIC_3	I never adapt university (-related) assignments prepared by ChatGPT or parts thereof, without citing ChatGPT as the source of information	
	ETHIC_4	I never copy university (-related) assignments prepared with ChatGPT and submit them for grading	
	ETHIC_5	I never cut out parts of a university(-related) assignment prepared by ChatGPT and re-draft them in a manner that suggests my point of view on the topic, even if it conflicts with the ideas contained in the ChatGPT-produced document	
	ETHIC_6	I consider preparing university(-related) assignments entirely with the ChatGPT to be unethical behavior.	
Perceived behavioral control	PBC_1	I have a high level of confidence in my technical proficiency in using ChatGPT effectively	[78,79]
	PBC_2	I am confident that I possess the necessary resources, knowledge, and skills to use ChatGPT effectively	
	PBC_3	I believe I have the ability to make the AI system behave in a way that aligns with my goals or intentions	
	PBC_4	I feel that I have full control over the process of using ChatGPT	

(continued on next page)

Table 1 (continued)

Construct	Item	Statement	Reference
Social/subjective norms/Social Influence (A - colleagues)	PBC_5	I feel that I have full control over both the process of using ChatGPT and the outcomes it produces	[67,79,80]
	SOCIALC_1	Colleagues who influence my behavior think that I should use ChatGPT	
	SOCIALC_2	Colleagues who are important to me think that I should use ChatGPT	
	SOCIALC_3	Colleagues take a positive view of me using ChatGPT	
	SOCIALC_4	Colleagues think that I should not use ChatGPT	
Social/subjective norms/Social Influence (B - professors)	SOCIALC_5	In general, my colleagues have supported the use of ChatGPT	[67,79,80]
	SOCIALP_1	My professors who influence my behavior recommend/advise me to use ChatGPT	
	SOCIALP_2	My professors recommend/advise that I should use ChatGPT	
	SOCIALP_3	My professors take a positive view of me using ChatGPT	
	SOCIALP_4	My professors recommend/advise that I should not use ChatGPT	
CHATGPT ADOPTION AND USE	SOCIALP_5	In general, the university has supported the use of ChatGPT	[67]
	EOU_1	Learning to use ChatGPT would be easy for me	
	EOU_2	I find it easy to navigate and use ChatGPT to accomplish tasks	
	EOU_3	My interaction with ChatGPT is clear and intuitive	
	EOU_4	I believe I can quickly become proficient at using ChatGPT	
Performance	EOU_5	ChatGPT feels user-friendly and easy to use	[67]
	PERF_1	I think that recommendations/ outputs provided by ChatGPT reflect my personalized preferences	
	PERF_2	I found the recommendations/ outputs provided by ChatGPT are a great match to my needs	
	PERF_3	I think the recommendations/ outputs provided by ChatGPT are accurate	
Intention to purchase	PERF_4	Recommendations/ outputs provided by ChatGPT are in general precise	Adapted from [72]
	INTENTPURCH_1	I intend to purchase a subscription for ChatGPT in the near future	
	INTENTPURCH_2	I am highly likely to buy a subscription for ChatGPT	
	INTENTPURCH_3	My intention to buy a subscription for ChatGPT is firm and unwavering	

Table 1 (continued)

Construct	Item	Statement	Reference
Intention to use	INTENTUSE_1	I am willing to use ChatGPT for university-related queries	[43,67]
	INTENTUSE_2	I am willing to make decisions based on the recommendations provided by ChatGPT	
	INTENTUSE_3	I intend to use ChatGPT for university(-related) assignments in the next 12 months	
Usefulness	USEFUL_1	Using ChatGPT enables me to accomplish tasks more quickly	[76,79]
	USEFUL_2	Using ChatGPT improves my performance at accomplishing tasks	
	USEFUL_3	Using ChatGPT for accomplishing tasks increases my productivity	
	USEFUL_4	Using ChatGPT enhances my effectiveness when it comes to accomplishing tasks	
	USEFUL_5	I find ChatGPT useful for accomplishing task	
Attitude towards use	ATU_1	Using ChatGPT was much better than what I expected	[76]
	ATU_2	I feel good using ChatGPT. The service provided by ChatGPT was better than what I expected	
	ATU_3	I feel comfortable with the services provided by ChatGPT	

and 54.81% respectively), and PL is mostly represented by bachelor's degree students (92.58%). Most respondents are aged from 21 to 25 (58.43%), but GE is characterized by a higher proportion of respondents aged from 26 to 30 (41.33%). The demographic characteristics of the respondents are as shown in Table 2.

Table 2

Summary of the respondents' demographic information by country (number and percentage).

	EE	GE	PL	RO	Total (%)
<i>Gender</i>					
Male	31	50	49	28	158 (45.93)
Female	21	21	32	107	181 (52.62)
Prefer not to say	0	3	0	0	3 (0.87)
<i>Level of education</i>					
Non-binary	1	1	0	0	2 (0.58)
Bachelor	17	55	75	61	208 (60.47)
Master	33	19	6	74	132 (38.37)
PhD	3	1	0	0	4 (1.16)
<i>Age</i>					
Below 21	4	7	28	48	87 (25.29)
From 21 to 25	39	29	51	82	201 (58.43)
From 26 to 30	6	31	1	3	41 (11.92)
From 31 to 40	4	8	1	2	15 (4.36)
TOTAL (%)	53 (15.41)	75 (21.80)	81 (23.55)	135 (39.24)	

4.2. Measurement model assessment

To examine students' attitude towards using ChatGPT for university assignments, first, we evaluated the reflective measurement model (Appendix B³). *Individual indicator reliability* was assessed by recommended individual item loadings with values greater than 0.708 [115]. In turn, Hair et al [115] and Hulland [117] suggest an outer loading greater than 0.4 is acceptable if the sample size is 200 or more. In our study, we take the cut-off value of individual indicator reliability equal to 0.6. Thus, in iteration 1 (Initial model) all elements of the model, except for three items of unethical ICT behavior construct (ETHIC_4, ETHIC_5 and ETHIC_6), as well as one item from social norms (colleagues) construct (SOCIALC_4), and social norms (Professors) (SOCIALP_4) construct, had an external load of less than 0.6 (see Appendix B). In iteration 2 (Final model), the exclusion of ETHIC_4, ETHIC_5 and ETHIC_6, SOCIALC_4 and SOCIALP_4 items led to an increase in the external load value of the remaining elements of the construct.

The *internal consistency reliability* is assessed using Cronbach's alpha (α) and Composite Reliability (CR). All constructs, with the exception of trust ($\alpha = 0.401$, AVE=0.463) demonstrate good internal consistency and good convergent validity, which explains more than 50% of the variance [115] (see Appendix C⁴). Regarding the trust construct, α and AVE values are considered acceptable based on the condition that, if the AVE is below 0.5, but CR exceeds 0.6, the construct's convergent validity is considered satisfactory (according to [118]). This criterion has been met, ensuring that convergent validity is achieved for each construct. Van Griethuijsen et al [119] posited that the occurrence of Cronbach's alpha coefficients falling below the acceptable threshold can be attributed to the limited number of items that contribute to the factors. Thus, the model can also be considered reliable.

Discriminant validity testing results confirm that all constructs demonstrating the values of the outer loading were above the values of all their cross-loadings on the other constructs [120] (see Appendix D⁵). The *Fornell-Larcker criterion* for assessing discriminant validity also confirms that discriminant validity is met since the entire value of the diagonal is greater than the value in its row and column (see Appendix E⁶). Our analysis also shows that all average correlations of the indicators across constructs (*HTMT ratio*) are below the threshold of 1.0 [121]. This indicates that the measurement model does not suffer from issues of discriminant validity (see Appendix F⁷). As presented in Appendix G⁸, all VIF values for the eleven latent variables were below the recommended threshold of 5.0 [116], ranging from 1.030 to 4.857. This demonstrates that multicollinearity is not an issue in the measurement model.

4.3. Structural model assessment

The evaluation of the structural model was conducted to assess the validity of the hypotheses proposed during the development of the research model. To investigate the statistical significance of the weights of the sub-constructs and the path coefficients (β), a bootstrapping procedure comprising 5000 iterations was executed. Examining the structural model was carried out to test the direct, indirect and moderating effects of the ChatGPT ethics perception, behavioral aspects of using ChatGPT, and technology adoption factors on students' intentions to use and purchase ChatGPT in support of educational activities (Table 3). Out of the 23 hypotheses, 20 were supported.

ChatGPT *ethical* aspects-related constructs demonstrated that 77.77% of the confirmed hypotheses were formulated in the study. Seven out of the nine hypotheses were supported. Privacy (H6: $\beta = 0.286$, $p < 0.01$), explainability (H1: $\beta = 0.251$, $p < 0.01$) and fairness (H5: $\beta = 0.150$, $p < 0.01$) were found to be strong predictors of trust. Transparency (H7: $\beta = 0.255$, $p < 0.01$) and accountability (H3: $\beta = 0.247$, $p < 0.01$) are strong predictors of fairness (H7: $\beta = 0.255$, $p < 0.01$). Transparency is also a predictor of explainability (H2: $\beta = 0.248$, $p < 0.01$). Our findings confirmed that trust significantly and positively affects performance when using ChatGPT (H8: $\beta = 0.381$, $p < 0.01$). However, accountability does not significantly predict explainability (H4: $\beta = 0.123$, $p = 0.101$), and trust does not significantly predict students' intentions to use ChatGPT for educational activities (H9: $\beta = 0.010$, $p = 0.864$).

Regarding *behavioral* aspects of using ChatGPT, all four hypotheses were supported. The social influence of university professors most strongly determines students' intentions to use ChatGPT (H13: $\beta = 0.219$, $p < 0.01$), followed by the social influence of university peers (H11: $\beta = 0.148$, $p < 0.05$) and perceived behavioral control of the process and outcomes of using ChatGPT (H10: $\beta = 0.122$, $p < 0.05$). The social influence of university professors was also revealed as only one behavioral predictor of the intention of students to purchase ChatGPT for engaging in educational activities (H12: $\beta = 0.200$, $p < 0.01$).

Regarding the *technology adoption* aspects of using ChatGPT, all eight (8) hypotheses were supported. Performance was identified as a strong predictor of the ease of use (H14: $\beta = 0.472$, $p < 0.01$) and proved able to predict significant attitude towards use of ChatGPT (H20: $\beta = 0.220$, $p < 0.01$). Ease of use, in turn, strongly positively impacts the perception of ChatGPT usefulness (H16: $\beta = 0.563$, $p < 0.01$) and significantly influences attitude towards use (H15: $\beta = 0.182$, $p < 0.01$). The strongest determinant of attitude towards the use of ChatGPT turned out to be usefulness (H18: $\beta = 0.429$, $p < 0.01$). At the same time, intentions to use ChatGPT for educational activities are first of all impacted by the perception of its usefulness (H19: $\beta = 0.200$, $p = 0.01$), and also influenced by student's attitude towards the use of ChatGPT (H20: $\beta = 0.163$, $p < 0.01$). Intentions to use along with the influence of university professors also determine the intentions to purchase ChatGPT (H21: $\beta = 0.325$, $p < 0.01$).

When constructing the initial structural model, the respondents' country and unethical ICT behavior factors were selected as relevant moderators. We also tested other demographic factors characterizing the sample, such as respondent's gender, age, and level of education. However, none of these factors demonstrated a significant moderating effect on any relationship. Therefore, these moderators were not included in the final model. Our results show that the respondent's Country negatively moderated the relationship between intentions to use and intentions to purchase ChatGPT (H23: $\beta = -0.123$, $p < 0.05$). We also revealed that unethical ICT behavior does not moderate the relationship between Trust and Intentions to use ChatGPT (H22: $\beta = -0.096$, $p = 0.053$).

R^2 for trust is equal to 0.292, for intentions to use is 0.381, and for intentions to purchase ChatGPT is equal to 0.211, and suggests that it is sufficient, particularly in the context of exploratory social science research [122]. These values of R^2 are sufficient for our research model estimation and they are presented in Appendix H⁹. The summary of the hypotheses' test results and path coefficient is presented in Table 3.

5. Discussion

This study provides a comprehensive understanding of students' intentions to use ChatGPT, one of the most widely used GenAI tools, for university assignments. Most existing studies have examined the adoption of ChatGPT or other GenAI tools in higher education using single-

³ Appendix B

⁴ Appendix C

⁵ Appendix D

⁶ Appendix E

⁷ Appendix F

⁸ Appendix G

⁹ Appendix H

Table 3

Summary of hypotheses test results and path coefficients.

H	Path	O	M	STDEV	T Statistics	P Values	Findings
<i>CHATGPT ETHICS</i>							
H1	EXPLAIN -> TRUST	0.251	0.253	0.056	4.510	0.000	Supported
H2	TRANSP -> EXPLAIN	0.248	0.250	0.069	3.583	0.000	Supported
H3	ACCOUNT -> FAIR	0.247	0.245	0.078	3.172	0.002	Supported
H4	ACCOUNT -> EXPLAIN	0.123	0.126	0.075	1.641	0.101	Not Supported
H5	FAIR -> TRUST	0.150	0.150	0.058	2.584	0.010	Supported
H6	PRIVACY -> TRUST	0.286	0.287	0.063	4.508	0.000	Supported
H7	TRANSP -> FAIR	0.255	0.260	0.075	3.377	0.001	Supported
H8	TRUST -> PERF	0.381	0.384	0.061	6.260	0.000	Supported
H9	TRUST -> INTENTUSE	0.010	0.007	0.056	0.171	0.864	Not Supported
<i>BEHAVIORAL ASPECTS OF USING CHATGPT</i>							
H10	PBC -> INTENTUSE	0.122	0.123	0.057	2.148	0.032	Supported
H11	SOCIALC -> INTENTUSE	0.148	0.154	0.059	2.518	0.012	Supported
H12	SOCIALP -> INTENTPURCH	0.200	0.203	0.061	3.294	0.001	Supported
H13	SOCIALP -> INTENTUSE	0.219	0.218	0.051	4.295	0.000	Supported
<i>CHATGPT ADOPTION AND USE</i>							
H14	PERF -> EASY OF USE	0.472	0.475	0.054	8.735	0.000	Supported
H15	PERF -> ATU	0.220	0.223	0.059	3.744	0.000	Supported
H16	EASY OF USE -> USEFUL	0.563	0.564	0.055	10.264	0.000	Supported
H17	EASY OF USE -> ATU	0.182	0.186	0.054	3.384	0.001	Supported
H18	USEFUL -> ATU	0.429	0.424	0.074	5.770	0.000	Supported
H19	USEFUL -> INTENTUSE	0.200	0.199	0.067	2.982	0.003	Supported
H20	ATU -> INTENTUSE	0.163	0.158	0.062	2.643	0.008	Supported
H21	INTENTUSE -> INTENTPURCH	0.325	0.324	0.053	6.143	0.003	Supported
<i>Moderating effect</i>							
H22	ETHIC x USEFUL -> INTENTUSE	-0.096	-0.090	0.049	1.937	0.053	Not Supported
H23	COUNTRY x INTENTUSE -> INTENTPURCH	-0.149	-0.150	0.048	3.099	0.002	Supported

theory models such as UTAUT [7,38], TAM [32,34], and TPB [35]. A few exceptions rely on other frameworks, for example SCT (Social Cognitive Theory) [55] and TCS (Technology Continuance Theory) [60]. Some studies extend one base model with additional constructs, such as Polypartis and Pahos [12], who extend UTAUT with anthropomorphism, trust, design novelty and institutional policy; Kalmus and Nikiforova [11], who extend IRT with selected TOE elements; and Jo and Bang [61], who propose TOEK, extending TOE with the concept of knowledge application.

Overall, most existing research relies on a single theory to study students' behaviour, attitudes and/or intentions to (continue to) use ChatGPT. This can be seen as limiting the scope of these studies to what is defined by the chosen theoretical lens. Exploring additional aspects often requires a completely new study, starting from theory development and moving to empirical application in a qualitative or quantitative design. By contrast, our integrated model considering ethical, behavioural, and technological drivers of adoption enables a broader view of the phenomenon within a single study, allowing researchers to investigate more intricate relationships between elements of multiple models, something that is not possible in single-theory approaches.

5.1. Factors influencing students' attitudes toward ChatGPT use

5.1.1. Ethical aspects

We found that students are concerned about the *ethical* aspects of ChatGPT, as implied by their experience of working with it. Numerous issues were raised by them, namely, erroneous information, lack of source transparency, reliability, and veracity, outdated/incomplete data, repetitiveness and vagueness, data privacy concerns, censorship, and biases, with the first three issues prevailing. These findings align with those of [123] who found the accuracy of ChatGPT to be around 60%, demanding careful assessment of its output before use, as well as with those of Rae et al [124], who pointed out that AI systems may unintentionally perpetuate and reinforce existing biases in the source material by analyzing vast but limited datasets. The respondents also pointed to the use of false and misleading references, which is also in line with Cooper [56], which makes students concerned about reliance on ChatGPT. However, this was not a concern for all respondents.

In more detail, we find that, generally, higher levels of explainability positively influence trust. Higher levels of explainability are generally associated with increased trust in AI systems, as shown by previous research suggesting that when users understand how an AI makes decisions, they are more likely to trust it [84]. Explainability provides users with insights into the decision-making processes of AI, reducing uncertainty and enhancing confidence in the system's outputs [45]. Transparency is another critical factor that positively influences trust. Transparent systems allow users to see and understand how decisions are made, which builds trust [88]. Our findings align with the literature that emphasizes the role of transparency in promoting user trust by making the system's operations visible and understandable [125]. Moreover, transparency also supports accountability, as it enables users to track and question AI decisions, ensuring that they adhere to expected standards [86]. Furthermore, our findings show that fairness is positively influenced by both accountability and transparency. When systems are transparent, students find it easier to assess their fairness, and accountability mechanisms ensure that any biases or unfair practices are identified and rectified [94]. This supports the notion that accountable and transparent systems are more likely to be perceived as fair, an aspect which is crucial for user acceptance and trust [36]. Trust is also significantly influenced by fairness and privacy considerations. Fair treatment by the AI and protection of user privacy contribute to a higher trust level [126]. Our research echoes the findings of previous studies, which show that users are more likely to trust and use AI systems that ensure fair treatment and that protect their privacy [127]. Trust is a key determinant of perceived performance. When users trust the system, they are more likely to view its performance favorably [104]. Interestingly, our study found that higher levels of accountability do not necessarily lead to increased explainability. This suggests that while a system may be accountable, it does not always mean that users will understand how it works. Accountability focuses on the ability to trace and justify decisions, whereas explainability emphasizes understanding how those decisions are arrived at. The distinction is crucial for designing AI systems that are both responsible and comprehensible [90]. Another notable finding is that higher levels of trust do not automatically translate into an increased intention to use the system. This indicates that while trust is important, other factors such as usability, perceived

usefulness, and individual preferences also play significant roles in the decision to adopt AI technology, thereby influencing user behavior [76]. This result aligns with findings from recent studies that were also not able to demonstrate the impact of trust on the adoption of ChatGPT within academia [33]. A possible explanation for the lack of any effect of trust on the intention to adopt is the novelty of using ChatGPT, as students might want first to learn how to use it and build a certain level of trust, and only afterward start using it for their academic endeavors.

5.1.2. Behavioral aspects

We also found that students' perceptions of their ability to control and navigate their interactions with ChatGPT significantly impact their intentions to use and purchase the system [31]. Control over interactions enhances students' confidence and comfort when it comes to using the technology, which is essential for their acceptance and continued use [105]. Moreover, the influence of social norms propagated by classmates and professors also plays a crucial role. When students observe their peers and instructors using and endorsing ChatGPT, they are more likely to develop a positive attitude towards the system and consider using it themselves. This social influence is consistent with the TPB, which posits that social norms and perceived behavioral control are key determinants of behavioral intentions [72]. Perceived performance positively influences ease of use and attitude towards use. When students perceive that ChatGPT performs well in terms of providing accurate and relevant information, they are more likely to find it easy to use and develop a favorable attitude towards its use [75]. The perceived ease of use reduces the effort required to learn and operate the system, enhancing user satisfaction and acceptance [76]. Usefulness is another critical factor that affects both ease of use and attitude towards use. When students find ChatGPT useful in their academic work, they are more inclined to use it regularly, which positively impacts their overall attitude towards the system [75]. This aligns with the TAM, which highlights the importance of perceived usefulness and perceived ease of use in determining users' attitudes and intentions [76]. Both usefulness and attitude towards use positively influence the intention to use ChatGPT. When students perceive the system as being beneficial and develop a positive attitude towards it, they are more likely to intend to use it consistently [76]. Finally, this intention to use translates into an intention to purchase. As students recognize the value and benefits of ChatGPT in their academic endeavors, they are more willing to invest in the system. The progression from intention to use to intention to purchase underscores the importance of initial user experience and satisfaction in driving long-term adoption and investment [109].

5.1.3. Technology adoption and use aspects

Our findings reveal a relationship between intention to use and intention to purchase ChatGPT. However, this is subject to negative moderation in terms of respondents' countries. Specifically, our findings indicate that in the case of respondents from PL and RO, the presence of an intention to use ChatGPT does not exert as significant an influence on their inclination to purchase the tool compared to students from EE and GE. While this can be due to several different factors, one of them, as our analysis implies, is the level of familiarity with ChatGPT. Our survey data (Sections 1 and 2) demonstrates variations across countries, with Estonian and German students exhibiting greater familiarity with ChatGPT, employing it more frequently for university assignments, and receiving more support from the university and individual teaching staff (i.e., in GE, followed by EE, the number of educators that not only do not prohibit, but even recommend its use, is highest among the analyzed countries). Institutional support likely contributes to the higher adoption rates and more diversified usage purposes observed among respondents from these countries, including advanced applications such as coding and programming tasks. However, it's crucial to acknowledge the potential influence of demographic factors on these usage patterns. In the Romanian and Polish samples, most of the respondents were from bachelor programs, whereas the sample from EE and GE was made up of

bachelor, master, and PhD students. Moreover, most Estonian respondents and one of the most popular groups of German respondents in the context of the field of study represented technology, namely, software engineering or computer science. While the former, diversity of the level of study, could suggest a plausible association between higher study levels and increased utilization of GenAI tools, as advanced students may have more extensive exposure and integration opportunities within their academic pursuits, it should be stressed that neither level of education nor the other two analyzed moderating factors, namely, gender and age, have demonstrated a moderating effect on any relationship. This can be caused by the makeup of the sample, which was heterogeneous in this regard. Consequently, we do not exclude the role of this factor, given that the field of study is likely to be a decisive factor in the context of the discussed usage pattern.

5.1.4. Moderating effect

Surprisingly, our results revealed that unethical ICT behavior does not moderate the relationship between trust and intentions to use ChatGPT. This is most likely because students perceive ChatGPT as a tool over which they have a greater degree of control compared to other ICT platforms, thus mitigating concerns about unethical behavior (as our results suggest). This contrasts with previous research that highlighted the significant impact of perceived unethical behavior on trust and technology adoption [44]. One potential explanation for this finding is that ChatGPT, as an AI-driven tool, allows for more personalized interactions and greater transparency in its decision-making processes, which can help alleviate concerns about unethical practices. Moreover, the perception of control over ChatGPT might stem from its interactive nature, where students can directly influence and guide the responses they receive, enhancing their sense of agency, and reducing their fear of unethical behavior. The sense of control is crucial in educational settings, where students value tools that they can manipulate to meet their specific needs and learning goals [105]. However, further research employing qualitative or mixed-method approaches could help unpack the underlying reasons behind this unexpected finding. Qualitative studies, such as in-depth interviews or focus groups, could provide nuanced insights into students' perceptions and experiences regarding ChatGPT, revealing why they might feel a higher degree of control over this tool compared to other ICT platforms. Mixed-method research could combine quantitative data with qualitative insights to offer a comprehensive understanding of how ethical considerations influence trust and usage intentions in different contexts [128].

5.2. Usage patterns and challenges regarding ChatGPT integration into academia

Our analysis of the ChatGPT usage patterns revealed that information retrieval, text summarization, text editing, such as rewriting/paraphrasing/improving, coding/programming, and language translation, extend the results of Mogavi et al [129], according to whom content creation and editing emerged as the most prevalent. Respondents also stressed that one of the challenges they have faced with ChatGPT is the inaccuracy and reliability concern, where, in several instances, the same incorrect answer can be repeated, even after users attempt to correct ChatGPT. This conflicts with the strength of self-improving capability as found by previous research [40]. Another significant challenge arises from the tool's dependence on precise prompts, where ChatGPT typically requires detailed explanations to fully grasp the context of a question. However, when questions are overly specific or intricate, the tool may struggle to generate accurate responses.

Some students face additional challenges in the realm of programming tasks, where ChatGPT exhibits limitations when tasked with handling more advanced or nuanced programming problems. Users have reported difficulties in debugging code, understanding complex coding functions, and encountering inaccuracies in solving tasks that

surpass basic programming logic. Some students reported challenges related to the data ingestion and processing capabilities of ChatGPT, where the tool may struggle to analyze and interpret large volumes of text effectively, an aspect that is crucial for conducting comprehensive academic research. This limitation, in turn, hinders their ability to obtain deep insights or thorough analyses from the tool, impacting the quality and depth of their academic work. Overall, these challenges highlight the need for students to navigate and mitigate the limitations of ChatGPT effectively, and to avoid over-relying on it. Developing strategies to address these challenges, such as refining query formulation techniques, seeking supplementary resources for specific tasks such as programming, and employing alternative tools for data analysis, can enhance students' interactions with ChatGPT and optimize its utility in their academic endeavors.

Finally, some critical components seen by respondents to be lacking that prevent them from working effectively with ChatGPT were identified, where prompt engineering and understanding of AI and algorithms, followed by the ability to verify the accuracy of the information, are seen to be the most dominant ones. This underscores the importance of users possessing prior knowledge and critical thinking skills to formulate queries effectively and further process information produced by ChatGPT. This is in line with the research by Yang et al [25], Miao & Holmes [4] and Dwivedi et al [130], in which critical thinking is seen to be negatively affected by ChatGPT, and is one of the major risks associated with its wide adoption, including in higher education, where educators are expected to be the ones who should find opportunities to develop critical thinking in the new setting. As such, it can become imperative for higher education institutions to educate students on the respective points, thereby allowing them to develop the necessary knowledge and skills that are crucial for effective GenAI adoption, an element that has become essential in the current educational domain.

5.3. Theoretical and practical implications

From a theoretical perspective, this study contributes in four main ways. First, it proposes and validates an integrated ethical-behavioural-technological model of generative-AI adoption that joins FATE with TAM, TPB, and UTAUT, showing that these frameworks can be fruitfully combined rather than applied in isolation. By incorporating ethical and behavioral considerations, the model extends beyond traditional technology acceptance factors, emphasizing the importance of fairness, accountability, transparency, and trust in AI adoption [94]. Whereas most prior research has applied TAM, UTAUT, or TPB in isolation [63,67], thereby limiting explanatory scope, the present study shows how ethical constructs drawn from the FATE framework interact with classical predictors of adoption such as perceived usefulness, ease of use, and intention to use.

Second, the study clarifies the role of ethical constructs in AI adoption by showing that transparency and accountability increase perceived fairness and explainability, and that privacy, fairness and explainability are strong predictors of trust in ChatGPT. The empirical validation of this model enhances our understanding of the drivers and barriers influencing GenAI adoption, demonstrating that ethical concerns are as critical as usability and performance in shaping students' willingness to engage with AI tools. Hence, this integration offers a more comprehensive perspective on GenAI use in higher education and extends the explanatory power of traditional technology acceptance theories [46, 71].

Third, the study refines existing acceptance models by demonstrating that trust significantly enhances perceived performance but does not directly predict intention to use once usefulness and attitudes are controlled for, suggesting that in the generative-AI context, trust functions primarily as a performance heuristic rather than a direct driver of adoption. While trust is often regarded as a direct antecedent of adoption intention [44], our results suggest that its role is more indirect: trust shapes perceptions of performance but does not directly predict

intention to use. This insight contributes to the ongoing debate about the complexity of trust in human-AI interaction [45,86].

Fourth, the study extends TPB and UTAUT-based research by showing that social norms from professors exert a stronger effect than peer norms or perceived behavioural control on both intention to use and intention to purchase ChatGPT, and that country context moderates the path from intention to use to intention to purchase. The cross-country results add an important comparative dimension, revealing that institutional support and students' familiarity with AI significantly moderate adoption. This underscores the influence of cultural and institutional contexts, which remain rather underexplored in the existing adoption literature [68,131].

From a practical standpoint, our findings have several implications for higher education institutions and policymakers. Because professors' social influence is the strongest predictor of both intention to use and intention to purchase ChatGPT [39], universities that want to steer AI use cannot rely only on generic institutional statements; they must actively engage and train teaching staff, as their explicit encouragement or discouragement is what most strongly shapes students' behaviour. Institutions should invest in faculty training to enhance AI literacy, equipping educators with the knowledge needed to integrate AI into teaching strategies while maintaining academic integrity. Encouraging AI-assisted learning through ethical frameworks can help universities strike a balance between leveraging AI's benefits and preventing its misuse.

Since ethical perceptions (explainability, privacy, fairness, and transparency) primarily act through trust and perceived performance rather than directly through intention to use, interventions that focus solely on "building trust in AI" are unlikely to be sufficient. Institutions need to pair ethical safeguards and transparency with concrete, high-quality use cases where the performance benefits of ChatGPT for learning are visible to students. Recent work has proposed structured frameworks for AI-augmented curriculum innovation [15,16,132] and provided empirical evaluations of generative models' performance in curriculum design [16]. These developments suggest that responsible adoption policies should encompass not only student use but also institutional strategies that harness AI to modernize teaching and learning at a systemic level. Thus, aligning perceived usefulness with ethical principles ensures that GenAI tools enhance learning outcomes while mitigating risks such as misinformation, plagiarism, and over-reliance on automation. Universities can therefore strengthen adoption by creating AI ethics committees, updating codes of conduct, and aligning their policies with international guidelines such as UNESCO's *AI and Education: Guidance for Policy-Makers* [133]. This would further support the responsible implementation of AI tools, ensuring compliance with data privacy regulations and addressing ethical dilemmas surrounding AI use in education.

The finding that intention to use translates into intention to purchase differently across countries indicates that adoption strategies and licensing decisions should be tailored to local digitalisation levels, economic constraints and institutional support, instead of assuming a one-size-fits-all European policy. Finally, students themselves highlight missing competencies such as prompt-engineering skills, understanding of AI limitations, and critical evaluation of AI-generated content. Universities interested in responsible AI integration should therefore embed AI literacy, critical thinking, and prompt-design training into curricula rather than treating ChatGPT as a purely "optional" tool [37,130]. Embedding AI literacy modules into curricula would help prevent over-reliance and misuse while equipping students with the skills needed to engage responsibly with GenAI [4]. Beyond the ethical, behavioral, and technological factors examined in our model, it is important to note that GenAI tools differ from traditional ICT systems in that they possess anthropomorphic features, such as conversational style and perceived social presence may enhance perceived trust and engagement among students, but may not overcome educators' algorithmic aversion or concerns about pedagogical alignment [11,12].

5.4. Limitations and future research

Like all empirical studies, this research is subject to several limitations that should be considered when interpreting the findings. These limitations also highlight promising avenues for future investigation.

First, the study focused exclusively on ChatGPT as the most prominent GenAI tool at the time of data collection [4,134]. However, other tools such as Grammarly AI, Character.AI, Bard, Brainly, and CourseHero are also widely adopted, particularly in educational settings. Miao & Holmes [4] report that researchers are developing education-specific AI models like “EdGPT.” While our empirical study focuses on ChatGPT, the designed model is tool-agnostic and can be adapted to assess the adoption of other GenAI tools by modifying measurement items accordingly. Future studies could apply the integrated model to a broader range of GenAI applications in order to compare adoption dynamics across platforms.

Second, the results represent a snapshot in time during a period of rapid technological evolution. ChatGPT’s functionalities, interfaces, and usage policies change frequently, while students’ familiarity, expectations, and risk perceptions also evolve. These dynamics may affect the stability of the constructs examined in our model. Longitudinal research is therefore needed to trace how trust, ethical concerns, and behavioral drivers of adoption shift over time. The developed model and supplementary materials enable replication, ensuring the study’s reproducibility. A longitudinal study in Dutch higher education [135] highlights that shifts in trust, emotional perceptions, and behavioral control influence AI usage over time.

Third, the study examined adoption exclusively from the perspective of students, thereby overlooking other crucial stakeholders in the higher education ecosystem, particularly faculty members and administrators. Educators play a decisive role in shaping students’ attitudes, while institutional policies influence acceptance and use. Future work should extend the analysis to these groups and explore how multi-stakeholder interactions facilitate or hinder responsible AI adoption in academic settings.

Finally, the study was conducted in four prominent European countries, i.e., Estonia, Germany, Poland, and Romania. While this cross-national scope provides valuable comparative insights, it does not capture other cultural, institutional, and regulatory contexts within and beyond Europe. Despite aligning with studies like Abdaljaleel et al [31] in Arab countries, the exclusion of diverse global regions limits broader applicability. Additionally, variations in sample homogeneity across the analyzed countries may affect the study’s ability to capture the full spectrum of factors shaping students’ AI adoption intentions. Hence, cross-cultural studies could further test the moderating effects of institutional support, cultural norms, and disciplinary traditions.

In sum, these limitations point to several directions for future research: expanding the range of GenAI tools studied, adopting longitudinal and mixed-methods designs, including additional stakeholders, and broadening the cultural scope of empirical investigations. Addressing these areas would not only validate and refine the integrated framework proposed in this study but also strengthen the evidence base for designing responsible and context-sensitive AI adoption strategies in higher education.

6. Conclusions

ChatGPT is still a new technology, but one which has been used extensively within academia in recent months. Students will continue to use it as they see the real value it may bring. However, the use of ChatGPT for academic endeavors comes with several challenges. This study highlighted the multifaceted nature of AI-enabled and smart technology adoption in education, emphasizing the need to consider the technological, behavioral, and ethical dimensions of GenAI adoption. The findings provide valuable guidance for effective implementation, ensuring that ChatGPT is not only user-friendly, but also ethically sound

and culturally sensitive. This research underscores the critical role of comprehensive frameworks, as it proposes an integrated model that incorporates TAM, UTAUT, TPB, and FATE in analyzing technology adoption, offering a rich understanding that informs future educational technology developments.

The insights gained pave the way for more targeted and effective strategies to promote AI adoption in education, addressing students’ practical needs and ethical concerns. Banning the use of ChatGPT is clearly not a feasible solution, as students will always find ways to use it. Therefore, promoting a balanced approach, where practical usability and ethical integrity go hand-in-hand, will ensure that AI tools such as ChatGPT make a positive contribution to the educational landscape. The comprehensive analysis serves as a valuable resource for understanding the complexities of adopting AI in education, guiding continued research and dialogue to refine these approaches and ensure effective integration.

This study’s findings emphasize the importance of considering both user experience and ethical implications, fostering an environment that promotes learning, innovation, and ethical responsibility. By prioritizing these considerations, educators can support students in utilizing advanced tools such as ChatGPT, enhance educational outcomes, and create a more inclusive learning environment. Ultimately, the integration of diverse theoretical frameworks provides a holistic understanding of AI adoption, offering a roadmap for thoughtful and effective implementation that benefits the entire educational ecosystem.

So what does this mean for higher education? Our results show that GenAI adoption in universities is not simply a matter of students trusting the technology or having access to it. Adoption is driven by a combination of ethical perceptions, perceived performance and usefulness, and, above all, the explicit stance taken by professors. If institutions merely ban ChatGPT or issue abstract warnings, they will be out of step with students’ actual practices and miss opportunities to shape them. By contrast, policies that invest in ethical safeguards, transparent communication, professor-led guidance, and AI-literacy training can harness students’ strong interest in ChatGPT while mitigating risks to academic integrity and learning.

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Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Nina Rizun: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ovidiu Niculae Bordean:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anastasija Nikiforova:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Ioana Natalia Beleiu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Aleksandra Revina:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

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Supplementary materials

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